

# Dongho Lee

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## Research Interests

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Symplectic Geometry, Symplectic Dynamics, Reeb Dynamics, Three-body Problem

## Education

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### Ph.D. in Mathematics

Department of Mathematics, Seoul National University, Seoul, Korea      Mar 2018 – Feb 2025

### Bachelor of Science in Mathematics

### Bachelor of Economics in Economics, Minor in Physics

College of Liberal Studies, SNU      Mar 2013 – Feb 2018

Graduated with Honors

## Research Experience

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### Postdoctoral Researcher

Center for Quantum Structures in Modules and Spaces (QSMS), SNU      May 2025 – Present

– Research on three-body problem and related topics.

Research Institute of Mathematics, SNU

Mar 2025 – Apr 2025

Postdoctoral Supervisor: Prof. Cheol-hyun Cho

– Research on periodic orbits in the three-body problem via symplectic geometry.

### Graduate Student Researcher

Department of Mathematics, SNU      Mar 2018 – Feb 2025

Ph.D Advisor: Prof. Otto van Koert

– Focus on global hypersurfaces of section and the three-body problem in symplectic and contact geometry.

## Publications

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1. **Conley-Zehnder Indices of Spatial Rotating Kepler Problem**      arXiv Preprint  
– Description of the moduli space and computing the Conley-Zehnder indices of periodic orbits of spatial rotating Kepler problem.
2. **Global Hypersurfaces of Section and the Spatial Kepler Problem**  
Advisor: Prof. Otto van Koert      Ph.D. Thesis, Dec 2024
3. **Global Hypersurfaces of Section for Geodesic Flows on Convex Hypersurfaces**  
with Sunghae Cho      *Archiv der Mathematik*, 31 Jul 2024  
– Constructed a global hypersurface of section for geodesic flows on convex hypersurfaces with isometric involution, generalizing Birkhoff annuli to higher dimensions.

## Invited Talks

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Sildes are available at my homepage.

1. **Hamiltonian Dynamics and Three-body Problem**  
Rookie's Pitch (at SNU), Seoul, Korea 14 Oct 2025
2. **Three-body Problem and Related Problems**  
QSMS 2025 Summer Workshop, Goseong, Korea 20–23 Aug 2025
3. **Periodic Orbits of Rotating Kepler Problem**  
Billiards and Quantitative Symplectic Geometry, Heidelberg, Germany 14 – 18 Jul 2025
4. **Closed orbits of the spatial rotating Kepler problem**  
2025 Korean Mathematical Society Spring Meeting, Daejeon, Korea 24 – 26 Apr 2025

## Personal Affiliations

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- Korean Mathematical Society**, Korea 2025 – Present
- QSMS-BK21 Symplectic Seminar**, Korea 2021 – Present

## Conference Participation

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- East Asian Symplectic Conference 2025** 25 – 29 Sep 2025  
Hokkaido University, Sapporo, Japan
- 2025 Symplectic Retreat** 23 – 26 Jan 2025  
Stanford Hotel, Tongyeong, Korea
- From Hamiltonian Dynamics to Symplectic Topology and Beyond** 3 – 7 Jun 2024  
A 2024 celebration of Claude Viterbo and his mathematics  
Institut Henri Poincaré, Paris, France
- Topology of 4-manifolds and Related Topics** 21 – 27 Jan 2024  
A conference in honor of Jongil Park's 60th birthday  
Ocean Suites Hotel, Jeju, Korea
- East Asian Symplectic Conference 2023** 29 Oct – 4 Nov 2023  
Ocean Suites Hotel, Jeju, Korea
- QSMS Workshop on Symplectic Geometry and Related Topics** 5 – 10 Feb 2023  
Ocean Suites Hotel, Jeju, Korea
- Workshop on Symplectic Dynamics** 27 Jun – 1 Jul 2022  
Instituto Superior Técnico, Lisboa, Portugal
- Symplectic Geometry and Beyond (Part I)** 7 – 10 Feb 2022  
Conference on the occasion of Yong-Geun Oh's 61st birthday  
Ocean Suite Hotel, Jeju, Korea
- Fukaya 60 Geometry and Everything** 17 – 22 Feb 2019  
A conference in honor of Kenji Fukaya's 60th birthday  
Kyoto University, Kyoto, Japan

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## Teaching Experience

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**Teaching and Course Assistant**, Seoul National University Mar 2018 – Feb 2025

### Major Courses

|                                                             |                          |
|-------------------------------------------------------------|--------------------------|
| <b>Algebraic Topology</b> (Graduate Course)                 | Fall 2019                |
| <b>Introduction to Topology</b>                             | Spring 2023, Spring 2020 |
| <b>Introduction to Differential Geometry</b>                | Spring, Fall 2021        |
| <b>Selected Topics Seminar</b> (College of Liberal Studies) | Spring 2023              |
| Probability in mathematical and philosophical viewpoints    |                          |

### General Education Courses

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|---------------------------------------------------------------------|-------------------------------------------------|
| <b>Head Teaching Assistant for Introductory Mathematics Courses</b> | Spring 2020                                     |
| Assisted in course planning and exam preparation                    |                                                 |
| <b>Engineering Mathematics</b>                                      | Fall 2024, Fall, Summer, Spring 2023, Fall 2020 |
| <b>Calculus for Life Science</b>                                    | Fall 2022, Spring 2021                          |
| <b>Fundamentals and Applications of Mathematics</b>                 | Spring 2022                                     |
| Calculus course for liberal arts students                           |                                                 |
| <b>Calculus Practice</b>                                            | Fall, Spring 2018                               |
| <b>Calculus</b>                                                     | Fall, Spring 2018                               |

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## Awards and Honors

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| <b>Merit-based Scholarship</b> , Department of Mathematics, Seoul National University  | 2018 |
| <b>National Scholarship</b> , Korean Student Aid Foundation                            | 2015 |
| <b>Merit-based Scholarship</b> , College of Liberal Studies, Seoul National University | 2013 |

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## References

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Prof. Otto van Koert  
Department of Mathematics, Seoul National University  
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Relationship: Ph.D. Advisor

Prof. Cheol-hyun Cho  
Department of Mathematics, POSTECH  
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Relationship: Postdoctoral Supervisor